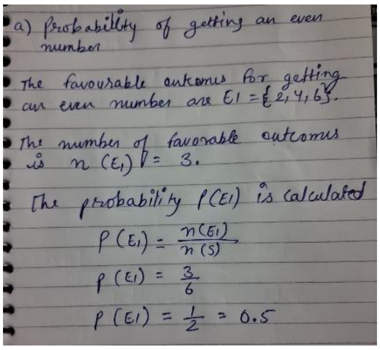
Probability basics

Q1. A die is rolled. What is the probability of getting:

(a) An even number

(b) A number greater than 4

Ans. (a)



a) Probability of getting an even number

The favourable outcomes for getting an even number are $E1 = \{2, 4, 6\}$.

The number of favorable outcomes is $n(E_1) = 3$.

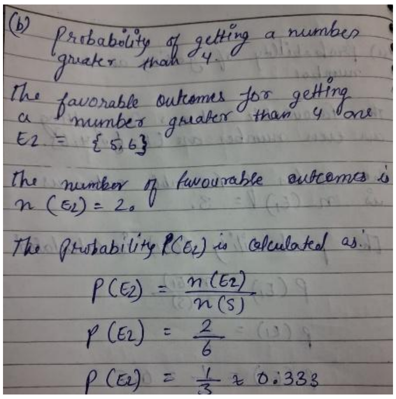
The probability $P(E1)$ is calculated

$$P(E_1) = \frac{n(E1)}{n(S)}$$

$$P(E1) = \frac{3}{6}$$

$$P(E1) = \frac{1}{2} = 0.5$$

(b)



(b) Probability of getting a number greater than 4.

The favorable outcomes for getting a number greater than 4 are $E2 = \{5, 6\}$

The number of favourable outcomes is $n(E_2) = 2$.

The probability $P(E_2)$ is calculated as:

$$P(E_2) = \frac{n(E2)}{n(S)}$$

$$P(E_2) = \frac{2}{6}$$

$$P(E_2) = \frac{1}{3} \approx 0.333$$

Q2. In a class of 50 students:

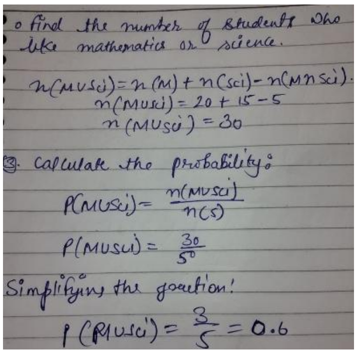
20 like Mathematics (M)

15 like Science (S)

5 like both subjects

What is the probability that a student chosen at random likes Mathematics or Science?

Ans. The probability that a student chosen at random likes Mathematics or Science is {3}{5} or 0.6 (60%).



• Find the number of students who like mathematics or science.

$$n(M \cup Sci) = n(M) + n(Sci) - n(M \cap Sci).$$

$$n(M \cup Sci) = 20 + 15 - 5$$

$$n(M \cup Sci) = 30$$

3. Calculate the probability:

$$P(M \cup Sci) = \frac{n(M \cup Sci)}{n(S)}$$

$$P(M \cup Sci) = \frac{30}{50}$$

Simplifying the fraction:

$$P(M \cup Sci) = \frac{3}{5} = 0.6$$

Q3. A bag has 3 red and 2 blue balls. If one ball is drawn randomly and is red, what is the probability that the next ball is also red (without replacement)?

Ans. Remaining red balls = 3 – 1 = 2

Remaining Total balls = 5 - 1 = 4

Calculation:

$$P\,(\text{Second red} \mid \text{First red}) = \frac{\text{Remaining Red}}{\text{Remaining Total}}$$

$$P = \frac{2}{4}$$

Final Ans :

The probability that the next ball is also red is $\frac{1}{2}$ or 0.5 (50%)

Q4. The population of a school is divided into 60% boys and 40% girls. If you want equal representation of both genders in the sample, which method should you use: Simple Random Sampling or Stratified Sampling? Why?

Ans. Sampling Method Selection

The correct method to use is Stratified Random Sampling.

Reasoning:

1. Group Division (Strata): This method allows you to divide the population into two specific groups (strata): Boys and Girls.
2. Guaranteed Representation: Unlike Simple Random Sampling, where the 60/40 split in the population might lead to a biased sample by chance, Stratified Sampling allows you to deliberately select an equal number (50% each) from both groups.
3. Accuracy: It ensures that the smaller group (girls) is not underrepresented, providing equal statistical weight to both genders.

Q5. The average height of 1000 students = 160 cm. A sample of 100 students shows an average height = 158 cm. Find the sampling error.

Ans. Sampling Error = 160 cm – 158 cm

Sampling Error = 2cm

Q6. The population mean salary is ₹50,000 with σ = ₹5,000. If we take a sample of 100 employees, what is the standard error of the mean (SEM)?

Ans6.

Ans6

Standard error of the mean (SEM)

① identify the given values.

- population standard deviation (σ): 5,000
- sample size (n): 100

② formula!

$$SEM = \frac{\sigma}{\sqrt{100}}$$

③ Calculation!

$$SEM = \frac{5000}{\sqrt{100}}$$

$$SEM = \frac{5000}{10}$$

$$SEM = ₹\,500$$

The Standard Error (₹500) represents the average amount that the sample mean is expected to vary from the actual population mean. A lower SEM indicates that the sample mean is a more accurate reflection of the true population mean.

Q7. In a group of 100 students:

40 like Cricket (C)

30 like Football (F)

10 like both Cricket and Football

Find the probability that a student likes at least one sport.

Ans7. The probability that a student likes at least one sport is 0.6 or 60%.

Find the number of students who like at least one sport

Formula! $n(C \cup F) = n(C) + n(F) - n(C \cap F)$

$$n(C \cup F) = 40 + 30 - 10$$

$$n(C \cup F) = 60$$

Calculate

$$P(C \cup F) = \frac{n(C \cup F)}{n(S)}$$

$$P(C \cup F) = \frac{60}{100}$$

Q8. From a deck of 52 cards, two cards are drawn without replacement. What is the probability that both are Aces?

Ans.

$P(\text{First card Ace}) = \frac{4}{52} = \frac{1}{13}$

$P(\text{second card Ace} \mid \text{first card Ace}) = \frac{3}{51} = \frac{1}{17}$

$$P(\text{Both Aces}) = \frac{4}{52} \times \frac{3}{51}$$

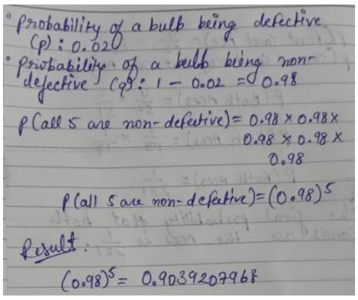
$$P(\text{Both Aces}) = \frac{1}{13} \times \frac{1}{17}$$

$$P(\text{Both Aces}) = \frac{1}{221}$$

The final probability that both cards are the Aces is $\frac{1}{221}$.

Q9. A factory produces bulbs with 2% defective rate. If 5 bulbs are chosen at random, what is the probability that all are non-defective?

Ans9.



- Probability of a bulb being defective $(p) : 0.02$
- Probability of a bulb being non-defective $(q) : 1 - 0.02 = 0.98$

$P(\text{all 5 are non-defective}) = 0.98 \times 0.98 \times 0.98 \times 0.98 \times 0.98$

$P(\text{all 5 are non-defective}) = (0.98)^5$

Result:

$(0.98)^5 = 0.9039207968$

The probability that all 5 bulbs are non-defective is approximately 0.9039 (or 90.39%).

Q10. Differentiate between discrete and continuous random variables with examples.

Ans. The Core Difference

- Discrete: Values you count. They are separate, distinct numbers (usually integers).
- Continuous: Values you measure. They can be any number (including decimals) within a range.

DRIVE LINK: https://drive.google.com/drive/folders/1ZlFjacMcn7A-Gi0Qmzbq18jmXQvYIuqH?usp=drive_link